Q1)

Code:

Terminal

```
FILENAME=sample.txt

create_file:
        echo "This is a sample file content" > $(FILENAME)

print_file:
        cat $(FILENAME)

clean:
        rm -f $(FILENAME)
```

Output:

Terminal

```
bodike@bodike-chaithali:~/LabActivity_07$ ls
bodike@bodike-chaithali:~/LabActivity_07$ mkdir B24CS013_01
bodike@bodike-chaithali:~/LabActivity_07$ cd B24CS013_01/
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ ls
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ vi makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ ls
makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ make create_file
echo "This is a sample file content" > sample.txt
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ ls
makefile  sample.txt
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ cat sample.txt
This is a sample file content
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ make print_file
cat sample.txt
This is a sample file content
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ ls
makefile  sample.txt
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ cat sample.txt
This is a sample file content
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ make clean
rm -f sample.txt
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_01$ ls
makefile
```

Q2)

Code:

Main.c

```c
#include<stdio.h>
#include"pallindrome.h"

int main () {
        int n;
        printf("Enter a number: ");
        if (scanf("%d",&n) != 1) return 1;

        if (is_pallindrome(n))
                printf("%d is a pallindrome.\n",n);
        else
                printf("%d is not a pallindrome.\n",n);
        return 0;
}
```

Pallindrome.h

```c
int is_pallindrome(int n);
~
~
```

pallindrome.c

```c
#include"pallindrome.h"

int is_pallindrome(int n) {
        int m=n,rev=0;
        if(m<0) m=-m;
        while(m>0) {
                rev=rev*10+(m%10);
                m/=10;
        }
        if(n<0) rev=-rev;

        return (rev == n);
}
~
```

Makefile

Terminal

```
CC = gcc
CFLAGS = -Wall -I.
OBJ = main.o pallindrome.o
TARGET = pallindrome_check

all: $(TARGET)

$(TARGET): $(OBJ)
        $(CC) $(CFLAGS) -o $(TARGET) $(OBJ)

main.o: main.c pallindrome.h
        $(CC) $(CFLAGS) -c main.c

pallindrome.o: pallindrome.c pallindrome.h
        $(CC) $(CFLAGS) -c pallindrome.c

clean:
        rm -f $(OBJ) $(TARGET)
```

Output:

Terminal

```
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01
bodike@bodike-chaithali:~/LabActivity_07$ mkdir B24CS013_02
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01  B24CS013_02
bodike@bodike-chaithali:~/LabActivity_07$ cd B24CS013_02/
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ vi main.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
main.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ vi pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
\main.c pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ vi pallindrome.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
main.c  pallindrome.c  pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ vi makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
main.c  makefile  pallindrome.c  pallindrome.h
```

```
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
main.c  makefile  pallindrome.c  pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ make
gcc -Wall -I. -c main.c
gcc -Wall -I. -c pallindrome.c
gcc -Wall -I. -o pallindrome_check main.o pallindrome.o
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ls
main.c  makefile       pallindrome_check  pallindrome.o
main.o  pallindrome.c  pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ./pallindrome_check
Enter a number: 121
121 is a pallindrome.
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ./pallindrome_check
Enter a number: -123
-123 is not a pallindrome.
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$ ./pallindrome_check
Enter a number: -121
-121 is a pallindrome.
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_02$
```

Q3)

Code:

Terminal

```
cmake_minimum_required(VERSION 3.10)

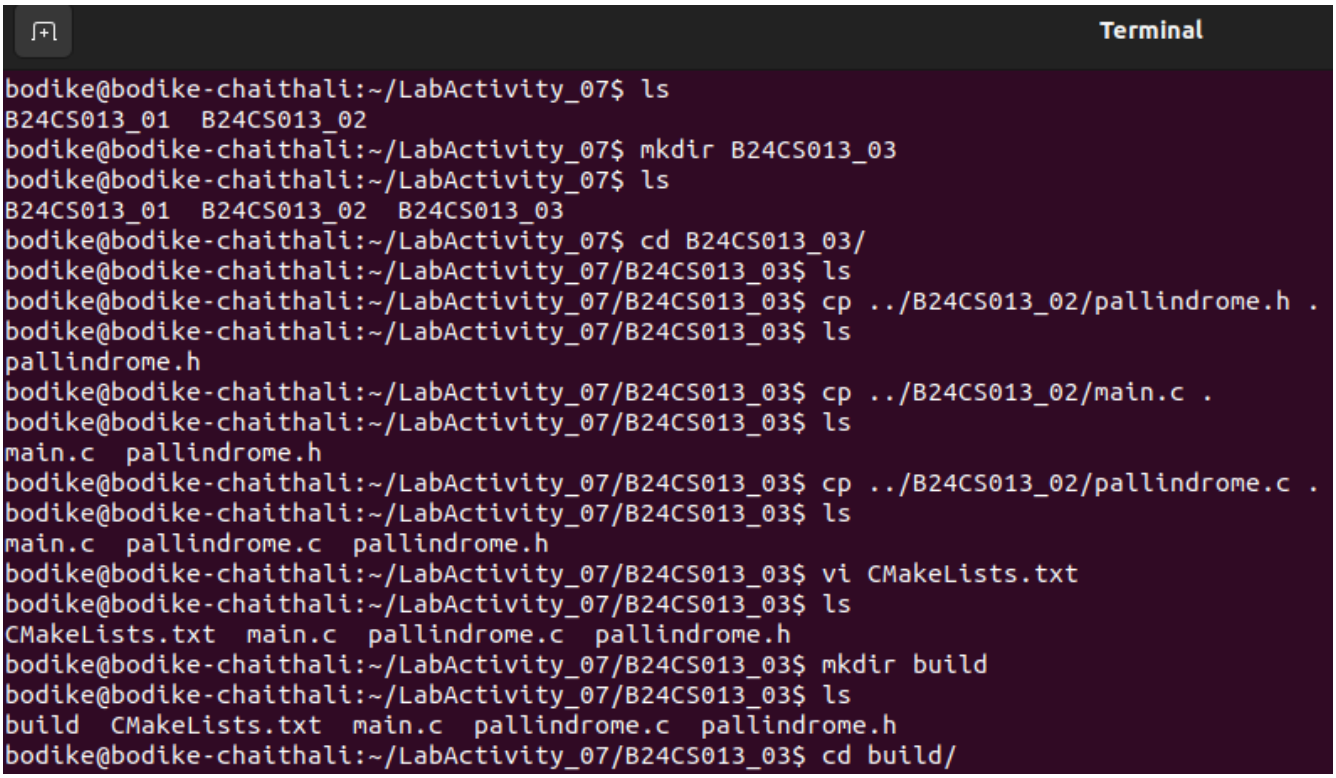
project(pallindrome_check C)

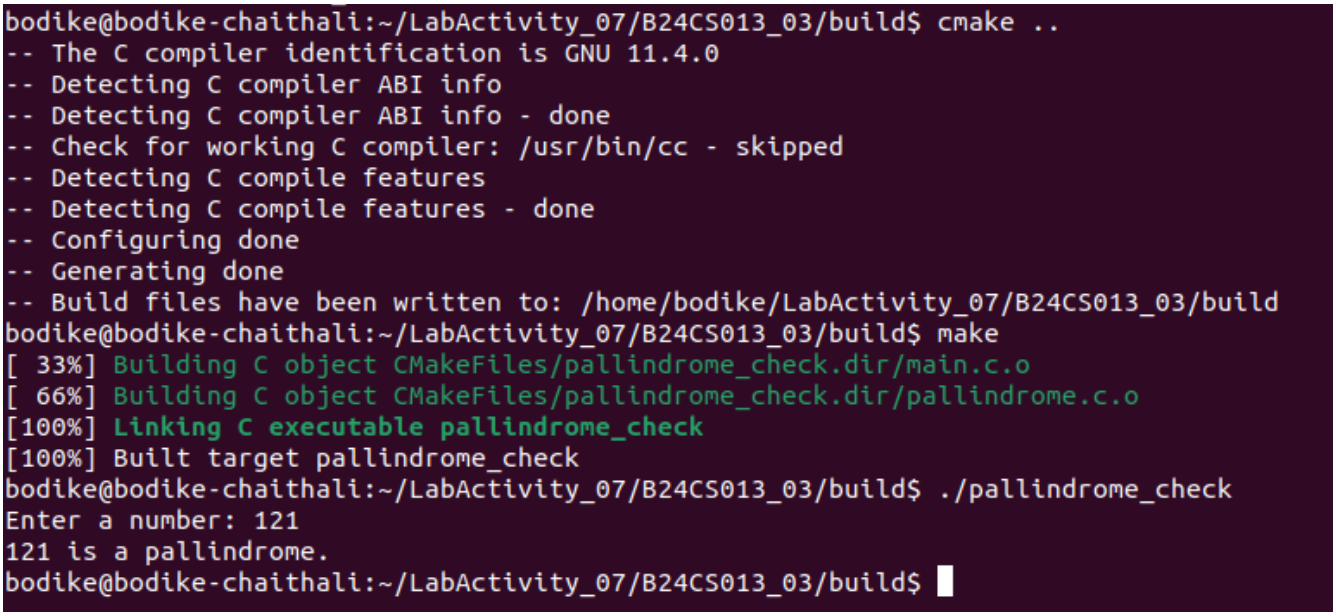
set(CMAKE_C_STANDARD 11)

set(SOURCES
        main.c
        pallindrome.c
)
add_executable(pallindrome_check ${SOURCES})

target_include_directories(pallindrome_check PRIVATE ${CMAKE_CURRENT_SOURCE_DIR})
```

Output:

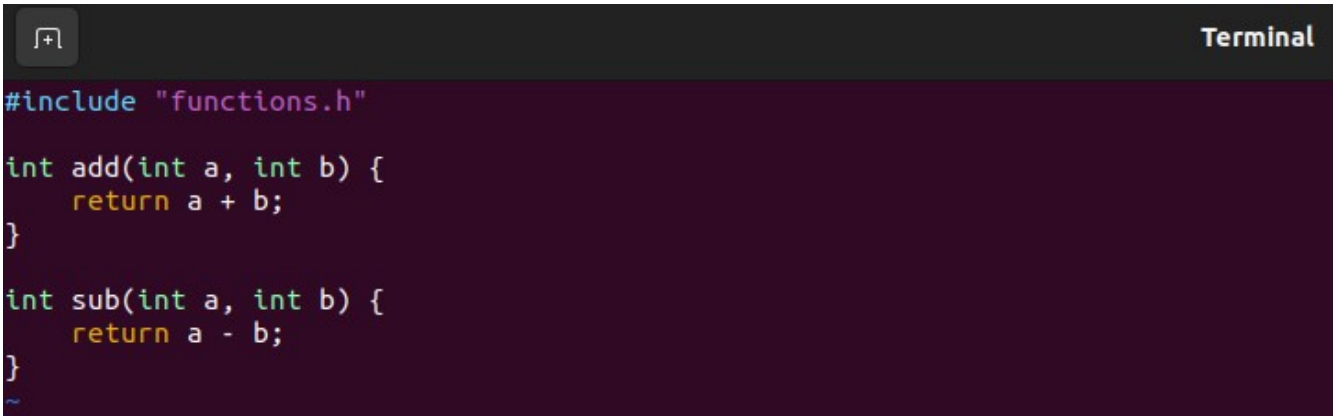

Terminal
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01 B24CS013_02
bodike@bodike-chaithali:~/LabActivity_07$ mkdir B24CS013_03
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01 B24CS013_02 B24CS013_03
bodike@bodike-chaithali:~/LabActivity_07$ cd B24CS013_03/
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ ls
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ cp ../B24CS013_02/pallindrome.h .
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ ls
pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ cp ../B24CS013_02/main.c .
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ ls
main.c pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ cp ../B24CS013_02/pallindrome.c .
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ ls
main.c pallindrome.c pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ vi CMakeLists.txt
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ ls
CMakeLists.txt main.c pallindrome.c pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ mkdir build
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ ls
build CMakeLists.txt main.c pallindrome.c pallindrome.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03$ cd build/


bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03/build$ cmake ..
-- The C compiler identification is GNU 11.4.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Configuring done
-- Generating done
-- Build files have been written to: /home/bodike/LabActivity_07/B24CS013_03/build
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03/build$ make
[33%] Building C object CMakeFiles/pallindrome_check.dir/main.c.o
[66%] Building C object CMakeFiles/pallindrome_check.dir/pallindrome.c.o
[100%] Linking C executable pallindrome_check
[100%] Built target pallindrome_check
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03/build$ ./pallindrome_check
Enter a number: 121
121 is a pallindrome.
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_03/build$

Q4)

Code:

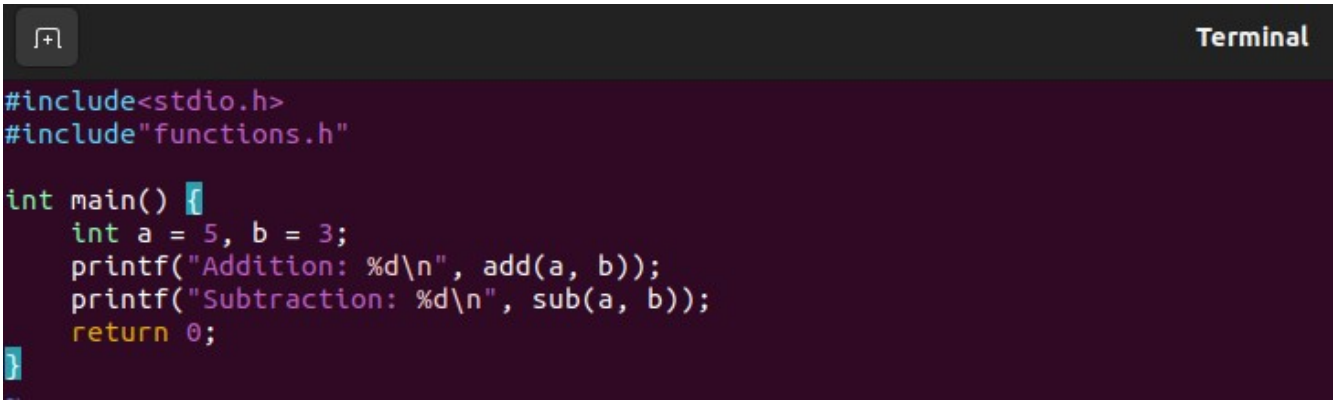
Functions.c



Functions.h



main.c



makefile

Terminal

```
CC = gcc
CFLAGS = -Wall -std=c11
OBJ = main.o functions.o
TARGET = program

$(TARGET): $(OBJ)
	$(CC) $(CFLAGS) -o $(TARGET) $(OBJ)
main.o: main.c functions.h
	$(CC) $(CFLAGS) -c main.c
functions.o: functions.c functions.h
	$(CC) $(CFLAGS) -c functions.c
clean:
	rm -f $(OBJ) $(TARGET)
```

Output:

Terminal

```
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01  B24CS013_02  B24CS013_03
bodike@bodike-chaithali:~/LabActivity_07$ mkdir B24CS013_04
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01  B24CS013_02  B24CS013_03  B24CS013_04
bodike@bodike-chaithali:~/LabActivity_07$ cd B24CS013_04/
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi functions.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
functions.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi functions.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
functions.c  functions.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi main.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
functions.c  functions.h  main.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
functions.c  functions.h  main.c  makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ make
gcc -Wall -std=c11 -c main.c
gcc -Wall -std=c11 -c functions.c
gcc -Wall -std=c11 -o program main.o functions.o
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ./program
Addition: 8
Subtraction: 2
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
functions.c  functions.h  functions.o  main.c  main.o  makefile  program
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ make clean
rm -f main.o functions.o program
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ ls
functions.c  functions.h  main.c  makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$
```

Q5)

Code:

run_primes.sh



prime.c

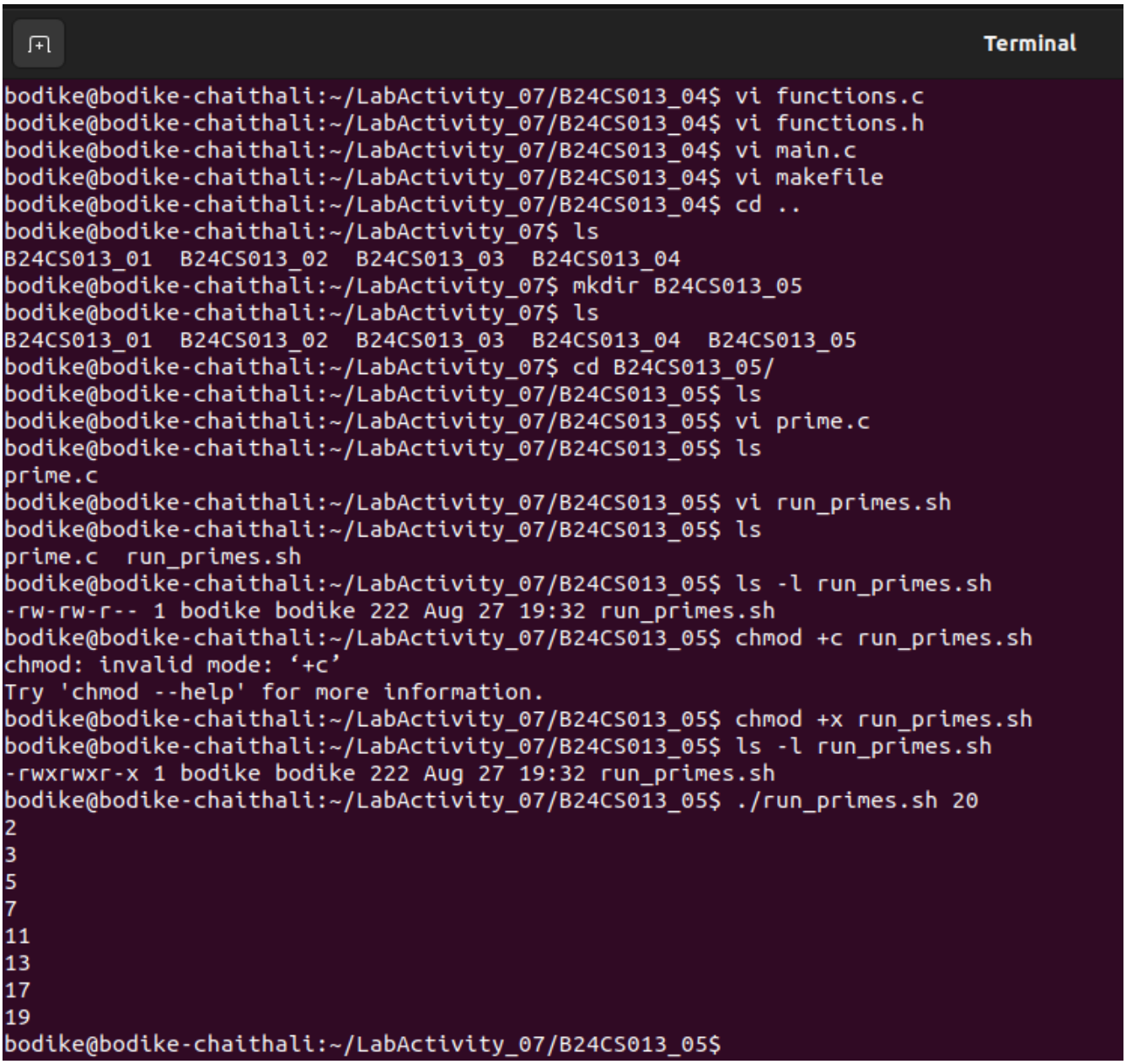
Terminal

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int isPrime(int n) {
    if (n <= 1) return 0;
    if (n == 2) return 1;
    if (n % 2 == 0) return 0;
    for (int i = 2; i < n/2 ; i ++) {
        if (n % i == 0) return 0;
    }
    return 1;
}

int main(int argc, char *argv[]) {
    if (argc != 2) {
        printf("Usage: %s <number>\n", argv[0]);
        return 1;
    }
    int num = atoi(argv[1]);
    if (isPrime(num)) {
        printf("%d\n", num);
    }
    return 0;
}
```

Output:


Terminal
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi functions.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi functions.h
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi main.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ vi makefile
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_04$ cd ..
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01 B24CS013_02 B24CS013_03 B24CS013_04
bodike@bodike-chaithali:~/LabActivity_07$ mkdir B24CS013_05
bodike@bodike-chaithali:~/LabActivity_07$ ls
B24CS013_01 B24CS013_02 B24CS013_03 B24CS013_04 B24CS013_05
bodike@bodike-chaithali:~/LabActivity_07$ cd B24CS013_05/
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ ls
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ vi prime.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ ls
prime.c
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ vi run_primes.sh
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ ls
prime.c run_primes.sh
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ ls -l run_primes.sh
-rw-rw-r-- 1 bodike bodike 222 Aug 27 19:32 run_primes.sh
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ chmod +c run_primes.sh
chmod: invalid mode: ‘+c’
Try 'chmod --help' for more information.
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ chmod +x run_primes.sh
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ ls -l run_primes.sh
-rwxrwxr-x 1 bodike bodike 222 Aug 27 19:32 run_primes.sh
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$ ./run_primes.sh 20
2
3
5
7
11
13
17
19
bodike@bodike-chaithali:~/LabActivity_07/B24CS013_05$